

PHYSIOLOGY 2017 EXAM

INTRO MODULE

1. Which one of the following is both a negative and positive feedback mechanism?

- A. Regulation of blood glucose
- B. Maintenance of calcium level in the ECF
- C. Hemostasis
- D. Control of body temperature
- E. Generation of action potential

Answer: C. Hemostasis

Explanation:

- Positive feedback: Platelet activation leads to more platelet recruitment and platelet plug formation.
- Negative feedback: Anticoagulant mechanisms prevent excessive clotting.
- A: Blood glucose uses primarily negative feedback.
- B: Calcium regulation uses primarily negative feedback.
- C: Hemostasis uses both.
- D: Body temperature is classic negative feedback.
- E: Action potential uses positive feedback but not both types.

2. Which cellular structure is NOT involved in protein synthesis?

- A. Rough ER
- B. DNA and RNA
- C. Ribosomes
- D. Nucleus and nucleoli
- E. None of the above

Answer: E. None of the above

Explanation:

- A: Rough ER has ribosomes that synthesize proteins.
- B: DNA holds genetic code; RNA carries it.
- C: Ribosomes are the direct site of translation.
- D: Nucleus/nucleoli are where transcription occurs and rRNA is made.
- E: All are involved.

3. The major components of the cell membrane are:

- A. Proteins and carbohydrates
- B. Proteins and lipids
- C. Lipids and carbohydrates
- D. Proteins and vitamins
- E. Carbohydrates and cholesterol

Answer: B. Proteins and lipids

Explanation:

- Cell membrane = Lipid bilayer + Proteins + Carbohydrates (minor).
- A: Carbohydrates are not major structural components.
- B: Correct.
- C: Missing proteins.
- D: Vitamins are not major components.

- E: Missing proteins.

4. Which one of the following does NOT belong to the cell membrane proteins?

- A. Ion channels
- B. Na^+ - K^+ ATPase
- C. Carrier proteins
- D. Nuclear transcription factors
- E. Adenylyl cyclase

Answer: D. Nuclear transcription factors

Explanation:

- A: Ion channels are membrane proteins.
- B: Na^+ - K^+ ATPase is a membrane pump.
- C: Carrier proteins are membrane transport proteins.
- D: Nuclear transcription factors are in the nucleus, not the membrane.
- E: Adenylyl cyclase is a membrane enzyme.

5. Which substances are transported by primary active transport?

- A. Na^+ , K^+ , Ca^{2+} and H^+
- B. Glucose and fructose
- C. Amino acids
- D. Hormones and neurotransmitters
- E. Free fatty acids

Answer: A. Na^+ , K^+ , Ca^{2+} and H^+

Explanation:

- Primary active transport directly uses ATP.
- Examples: Na^+/K^+ -ATPase, Ca^{2+} -ATPase, H^+ -ATPase, H^+/K^+ -ATPase.
- B: Glucose uses facilitated diffusion or secondary active transport.
- C: Amino acids use secondary active transport.
- D: Hormones/neurotransmitters are not transported this way.
- E: Fatty acids diffuse or use transport proteins.

6. Which statement is correct?

- A. Lipid bilayer surfaces facing ECF/ICF are hydrophobic
- B. Polar molecules easily pass through lipid bilayer
- C. Protein channels within lipid bilayer allow passage of ions
- D. Polar heads face each other in the interior
- E. None

Answer: C. Protein channels within the lipid bilayer allow passage of ions

Explanation:

- A: Polar heads face ECF/ICF; hydrophobic tails face inward.
- B: Polar/charged substances cannot easily cross the hydrophobic core.
- C: Ion channels provide hydrophilic pathways.
- D: Hydrophobic tails face each other internally.
- E: C is correct.

7. Which is WRONG about Na^+ - K^+ ATPase?

- A. It has four binding sites interiorly
- B. Contributes to resting membrane potential
- C. Hydrolyzes ATP
- D. Transports 3 Na^+ inward and 2 K^+ outward

E. Has K^+ binding sites on ECF face

Answer: D. It transports 3 Na^+ inward and 2 K^+ outward

Explanation:

- The Na^+/K^+ pump moves 3 Na^+ OUT and 2 K^+ IN.
- D: Direction is reversed — this is the wrong statement.
- A: Imprecise wording but multiple binding sites exist.
- B: It is electrogenic and contributes to resting potential.
- C: It is an ATPase.
- E: K^+ binds on the extracellular side.

8. Which substances readily diffuse through the lipid bilayer?

A. O_2 and CO_2

B. Water

C. Na^+ and K^+

D. Glucose

E. Urea

Answer: A. O_2 and CO_2

Explanation:

- Small nonpolar molecules cross readily.
- A: O_2/CO_2 are lipid-soluble gases.
- B: Water crosses mainly through aquaporins.
- C: Charged ions cannot cross the hydrophobic core.
- D: Glucose is large and polar.
- E: Urea is polar and crosses less readily.

9. Glucose and amino acids are absorbed from the GIT via:

A. Simple diffusion

B. Osmosis

C. Secondary active transport

D. Primary active transport

E. Vesicular transport

Answer: C. Secondary active transport

Explanation:

- Glucose and amino acids use Na^+ -dependent cotransport.
- A: Not the major mechanism.
- B: Osmosis transports water.
- C: Na^+ gradient drives glucose/amino acid uptake.
- D: Na^+/K^+ pump provides gradient indirectly.
- E: Not the normal mechanism.

10. Which electrolytes are present at a higher amount in ECF than ICF?

A. Na^+ , Cl^- , K^+ and HCO_3^-

B. K^+ , Cl^- , SO_4^{2-} and HCO_3^-

C. Na^+ , Cl^- , HCO_3^- and Ca^{2+}

D. K^+ , SO_4^{2-} , Mg^{2+} and PO_4^{3-}

E. Ca^{2+} , K^+ and Na^+

Answer: C. Na^+ , Cl^- , HCO_3^- and Ca^{2+}

Explanation:

- ECF: Na^+ , Cl^- , HCO_3^- , Ca^{2+}

- ICF: K^+ , Mg^{2+} , phosphate, proteins
- A: K^+ is predominantly ICF.
- B: K^+ and sulfate are not predominantly ECF.
- C: Correct.
- D: These are predominantly intracellular.
- E: K^+ is predominantly intracellular.

11. Which represents a symport carrier?

- A. Glucose facilitated diffusion
- B. Na^+ - K^+ pump
- C. Na-glucose cotransporter
- D. O_2 and CO_2 transport
- E. Osmosis

Answer: C. Na-glucose cotransporter

Explanation:

- Symport = two substances move in the same direction.
- A: Usually uniport.
- B: Antiport-like exchange (Na^+ out/ K^+ in).
- C: Na^+ and glucose move together into the cell.
- D: Diffusion, not symport.
- E: Water movement, not carrier symport.

12. Which proteins make up tight junctions?

- A. Connexins
- B. Claudin
- C. Cadherins
- D. Selectin
- E. Integrins

Answer: B. Claudin

Explanation:

- A: Connexins form gap junctions.
- B: Claudins are major tight-junction proteins.
- C: Cadherins form adherens junctions/desmosomes.
- D: Selectins are cell adhesion molecules.
- E: Integrins are cell-ECM adhesion proteins.

13. Cell bodies of unipolar neurons are located in:

- A. Anterior horn
- B. Retina and inner ear
- C. Lateral horn
- D. Posterior horn
- E. Dorsal root ganglia

Answer: E. Dorsal root ganglia

Explanation:

- A: Somatic motor neuron cell bodies.
- B: Specialized sensory neurons, not classic pseudounipolar.
- C: Autonomic preganglionic neurons.
- D: Sensory processing/interneurons.
- E: Dorsal root ganglia contain pseudounipolar sensory neurons.

14. Identify the correct statement.

- A. Motor neurons transmit impulses periphery to CNS
- B. Microglia provide nutrients to neurons
- C. Neuroglia are excitable like neurons
- D. CNS and PNS myelin are chemically identical
- E. Ependymal cells are nutritive cells in CNS

Answer: E. Ependymal cells are nutritive cells in the CNS

Explanation:

- A: Motor neurons carry impulses CNS to periphery.
- B: Microglia are immune cells.
- C: Neuroglia do not generate action potentials like neurons.
- D: CNS myelin = oligodendrocytes; PNS myelin = Schwann cells.
- E: Best available answer but poorly worded; ependymal cells line ventricles and help produce CSF.

15. Which is WRONG about the neuromuscular junction?

- A. Neurotransmitter is always acetylcholine
- B. Postsynaptic receptor is always nicotinic
- C. EPSP is always produced
- D. EPSP initiates contraction
- E. None is wrong

Answer: E. None of the above is wrong

Explanation:

- At the skeletal neuromuscular junction:
 - o A: Neurotransmitter = ACh.
 - o B: Receptor = nicotinic Nm.
 - o C: ACh produces depolarizing end-plate potential.
 - o D: End-plate potential initiates muscle action potential and contraction.
 - o E: Correct.

24. Which is parasympathomimetic?

- A. Propranolol
- B. Curareform
- C. Atropine
- D. Adrenaline
- E. Pilocarpine

Answer: E. Pilocarpine

Explanation:

- A: Propranolol is a beta-blocker, sympatholytic.
- B: Curareform is a neuromuscular blocker.
- C: Atropine is a muscarinic antagonist, parasympatholytic.
- D: Adrenaline is sympathomimetic.
- E: Pilocarpine is a muscarinic agonist, parasympathomimetic.

25. Mechanism of action of ACh when it binds M1 is via:

- A. DAG and IP₃
- B. Opening K⁺ channels
- C. Activation of G-protein
- D. Opening Na⁺ channels

E. Opening Ca^{2+} channels

Answer: A. Formation of DAG and IP_3

Explanation:

- $\text{M1} = \text{Gq} \rightarrow \text{PLC} \rightarrow \text{IP}_3 + \text{DAG} \rightarrow \text{increased intracellular } \text{Ca}^{2+}$.
- A: Correct.
- B: K^+ channels are characteristic of some Gi -coupled muscarinic effects, especially M_2 .
- C: M_1 does activate a G-protein, but A is the more specific downstream mechanism.
- D: Not the principal mechanism.
- E: Ca^{2+} rises mainly through IP_3 -mediated intracellular release.

26. Which is NOT a sympathomimetic drug?

- A. Epinephrine
- B. Norepinephrine
- C. Methoxamine
- D. Carbachol
- E. Dopamine

Answer: D. Carbachol

Explanation:

- A: Epinephrine is an alpha and beta agonist.
- B: Norepinephrine is an alpha and beta1 agonist.
- C: Methoxamine is an alpha1 agonist.
- D: Carbachol is a cholinergic agonist, parasympathomimetic.
- E: Dopamine is a dopaminergic agonist; at higher concentrations also has beta1/alpha1 effects.

27. Neurotransmitter to sweat glands is:

- A. Epinephrine
- B. Norepinephrine
- C. Acetylcholine
- D. Dopamine
- E. Serotonin

Answer: C. Acetylcholine

Explanation:

- Important exception: Most sympathetic postganglionic neurons release norepinephrine, BUT sympathetic fibers to eccrine sweat glands release ACh.
- A: Incorrect.
- B: Typical sympathetic neurotransmitter, but not for eccrine sweat glands.
- C: Correct.
- D: Incorrect.
- E: Incorrect.

28. Function of T-tubules in skeletal muscle:

- A. Store Ca^{2+}
- B. Conduct depolarization to SR
- C. Contain postsynaptic receptors
- D. Store neurotransmitters
- E. Allow Ca^{2+} entry into cell

Answer: B. Conduct depolarization waves to the sarcoplasmic reticulum

Explanation:

- A: SR stores Ca^{2+} .
- B: T-tubules carry the action potential deep into the muscle fiber.
- C: Nicotinic receptors are at the neuromuscular junction.
- D: Neurotransmitter is stored in presynaptic vesicles.
- E: Their primary role is electrical conduction, not Ca^{2+} entry.

29. Which does NOT receive dual autonomic innervation?

- A. Heart
- B. Blood vessels
- C. Liver
- D. Stomach
- E. Pancreas

Answer: B. Blood vessels

Explanation:

- Most blood vessels receive predominantly sympathetic autonomic innervation.
- A: Heart receives sympathetic and parasympathetic.
- B: Blood vessels are mostly sympathetic; many have little or no direct parasympathetic innervation.
- C: Liver has autonomic regulation, though the pattern is organ-specific.
- D: Stomach receives sympathetic and parasympathetic.
- E: Pancreas receives autonomic innervation from sympathetic and parasympathetic pathways.

30. Which cranial nerve is stimulatory to salivary glands?

- A. Oculomotor
- B. Facial
- C. Vagus
- D. Trigeminal
- E. Hypoglossal

Answer: B. Facial nerve (CN VII)

Explanation:

- CN VII parasympathetically stimulates the submandibular and sublingual glands.
- A: CN III goes to the eye.
- B: CN VII stimulates salivary secretion.
- C: CN X goes to thoracic/abdominal viscera.
- D: CN V carries sensory fibers and some motor functions but is not the parasympathetic secretomotor nerve.
- E: CN XII controls tongue movement.
- Also remember: CN IX stimulates the parotid gland.

31. Respiratory function of the respiratory system is:

- A. Gas conduction
- B. Phonation
- C. Maintenance of pH
- D. Excretory function
- E. BP regulation

Answer: C. Maintenance of pH

Explanation:

- CO_2 elimination by the lungs is an important component of acid–base regulation.

- A: Gas conduction is a mechanical/conducting function.
- B: Phonation is a specialized function of the larynx.
- C: Ventilation regulates PaCO_2 and therefore pH.
- D: CO_2 excretion occurs, but maintenance of pH is the best physiologic function among these.
- E: The respiratory system can influence BP, but it is not its primary respiratory function.

32. The work of breathing decreases when:

- A. Lung compliance increases
- B. Subject exercises
- C. Respiratory rate increases with constant minute volume
- D. Person lies down
- E. FRC increases

Answer: A. Lung compliance increases

Explanation:

- Higher compliance means less pressure is required to expand lungs, so less elastic work.
- A: Correct.
- B: Exercise increases ventilatory work.
- C: With constant minute ventilation, increasing rate can increase work.
- D: Supine position can alter respiratory mechanics and often increases work of breathing.
- E: Increased FRC does not necessarily decrease total work.

33. Wrong about vital capacity:

- A. Air expelled from full inspiration to full expiration
- B. Reduced as one grows older
- C. Greater in men than women of same age/height
- D. Not related more to total body mass than lean body mass
- E. Sum of inspiratory and expiratory reserve volumes

Answer: E is the wrong statement

Explanation:

- Vital capacity = IRV + TV + ERV.
- E is incomplete because it omits tidal volume.
- A: Correct definition of VC.
- B: VC generally decreases with aging.
- C: Average VC is greater in men of the same age/height.
- D: The wording is poor, but body composition/body size affects lung volumes.
- E: Should be IRV + TV + ERV.

34. Not true of respiratory system function:

- A. Gas exchange
- B. Does not affect venous return
- C. Regulates BP
- D. Regulates acid-base balance
- E. Produces surfactant

Answer: B. Does not affect venous return

Explanation:

- Respiration does affect venous return through changes in intrathoracic pressure.
- A: Gas exchange is true.
- B: Respiratory pressure changes influence venous return, so this is false.

- C: Respiratory system contributes to BP regulation.
- D: Controls CO₂ and therefore acid-base status.
- E: Type II pneumocytes produce surfactant.

35. Minute ventilation is less than alveolar ventilation.

- A. True
- B. False

Answer: B. False

Explanation:

- Minute ventilation = $V_T \times \text{respiratory rate}$.
- Alveolar ventilation = $(V_T - V_D) \times \text{respiratory rate}$.
- Because $V_D > 0$, alveolar ventilation < minute ventilation.
- So the statement is false.

36. Lack of surfactant in infant RDS results in:

- A. Increased lung compliance
- B. Stabilization of alveolar volume
- C. Increased recoil force due to increased surface tension
- D. Reduced A-a O₂ difference
- E. Decreased filtration forces

Answer: C. Increased recoil force of lungs due to increased alveolar surface tension

Explanation:

- Surfactant decreases surface tension and increases compliance.
- Without surfactant: increased surface tension, increased elastic recoil, decreased compliance, alveolar collapse.
- A: Compliance decreases.
- B: Surfactant stabilizes alveoli; deficiency promotes collapse.
- C: Correct.
- D: Gas exchange worsens and A-a O₂ gradient increases.
- E: Not the primary effect.

37. Volumes/capacities that cannot be measured by direct spirometry:

- A. RV, ERV, IC
- B. RV, FRC, TLC
- C. IRV, RV, VC
- D. TLC, RV, ERV
- E. TV, IRV, IC

Answer: B. RV, FRC and TLC

Explanation:

- Any volume containing residual volume cannot be directly measured by spirometry.
- A: ERV and IC can be measured.
- B: RV, FRC, TLC require another technique.
- C: IRV and VC can be measured.
- D: ERV can be measured.
- E: All listed can be measured directly.
- Spirometry cannot measure RV, therefore cannot directly measure FRC or TLC.

38. Total lung capacity consists of:

- A. VC + RV
- B. RV + VC

- C. FRC + VC
- D. TV + FRC
- E. TV + RV

Answer: A. Vital capacity + residual volume

Explanation:

- $TLC = VC + RV$.
- $VC = IRV + TV + ERV$.
- **A: Correct.**
- **B: Mathematically identical to A, duplicate answer.**
- C: FRC + VC is not TLC.
- $D: TV + FRC = ERV + TV + RV$.
- E: Does not include IRV/ERV.
- Question has duplicate correct choices A and B.

39. All occur during expiration EXCEPT:

- A. Inspiratory muscles relax and rib cage descends
- B. Thoracic cavity volume decreases
- C. Intrapulmonary volume decreases
- D. Intrapulmonary pressure rises above atmospheric pressure
- E. Gases flow out against pressure gradient

Answer: E. Gases flow out against the pressure gradient

Explanation:

- Gases flow down a pressure gradient, not against it.
- During quiet expiration: decreased thoracic volume, increased alveolar pressure, air flows out.
- A: Correct.
- B: Correct.
- C: Correct.
- D: Alveolar pressure becomes slightly positive relative to atmospheric pressure.
- E: Air flows from higher pressure to lower pressure, so this is false.

48. Part of renal tubules where nutritionally important substances are reabsorbed:

- A. Collecting duct
- B. Proximal tubule
- C. Distal tubule
- D. Loop of Henle
- E. Urinary bladder

Answer: B. Proximal tubule

Explanation:

- The proximal tubule reabsorbs large amounts of glucose, amino acids, Na^+ , HCO_3^- , water, and other nutrients.
- A: Mainly final regulation of water/electrolytes.
- B: Correct.
- C: Fine-tuning of electrolytes.
- D: Major role in creating medullary osmotic gradient.
- E: Not a renal tubule.

49. Renal threshold for glucose is equal to:

- A. 340 mg/mL

- B. 200 mg/dL
- C. 125 mL/min
- D. 80–120 mg/dL
- E. 40 mg/dL

Answer: B. 200 mg/dL

Explanation:

- Renal threshold for glucose is approximately 180–200 mg/dL.
- A: Wrong unit/magnitude.
- B: Correct.
- C: This is a flow unit and relates approximately to GFR.
- D: Normal fasting plasma glucose range is not renal threshold.
- E: Too low.

50. Tubules on which ADH and aldosterone act to promote water-electrolyte reabsorption:

- A. Distal tubules and collecting duct
- B. Proximal tubules
- C. Loop of Henle
- D. Glomerulus
- E. Afferent/efferent arterioles

Answer: A. Distal tubules and collecting duct

Explanation:

- ADH acts strongly on collecting ducts; aldosterone acts mainly on late distal tubule/collecting duct.
- A: Correct.
- B: PCT reabsorption is largely not directly controlled by these hormones.
- C: ADH/aldosterone are not the primary regulators of the loop.
- D: Filtration occurs here.
- E: These are blood vessels, not renal tubules.

51. Which factor increases GFR?

- A. Afferent constriction
- B. Efferent vasodilation
- C. Decreased filtration pressure
- D. Congestive heart failure
- E. Increased renal blood flow

Answer: E. Increased renal blood flow

Explanation:

- A: Decreases renal blood flow and generally decreases GFR.
- B: Efferent dilation decreases glomerular capillary pressure and decreases GFR.
- C: Directly decreases GFR.
- D: Usually decreases renal perfusion/GFR.
- E: Increased renal blood flow can increase glomerular filtration, depending on vascular resistance.

52. Renal tubule involved in formation of concentrated urine:

- A. Collecting duct
- B. Proximal tubule
- C. Distal tubule
- D. Loop of Henle

E. Glomerulus

Answer: D. Loop of Henle

Explanation:

- The loop of Henle establishes the medullary osmotic gradient through countercurrent multiplication.
- A: Collecting duct is where ADH allows water reabsorption and final urine concentration.
- B: Major bulk reabsorption but does not create the medullary gradient.
- C: Fine regulation.
- D: Critical for establishing the concentration gradient.
- E: Filtration occurs here.

53. Which is TRUE regarding control of GFR and RBF?

- A. Strong sympathetic activation causes marked efferent constriction
- B. Angiotensin II causes efferent arteriolar constriction
- C. Angiotensin II increases peritubular capillary flow
- D. ANP decreases GFR
- E. Endothelin increases GFR

Answer: B. Angiotensin II causes efferent arteriolar vasoconstriction

Explanation:

- A: Sympathetic stimulation primarily causes renal arteriolar vasoconstriction, particularly affecting afferent flow.
- B: Angiotensin II preferentially constricts the efferent arteriole, helping maintain glomerular pressure.
- C: Ang II tends to decrease renal/peritubular capillary blood flow.
- D: ANP generally increases GFR.
- E: Endothelin causes vasoconstriction and generally decreases GFR.

54. Which factor is NOT in favor of GFR?

- A. Increased glomerular capillary hydrostatic pressure
- B. Decreased capsular hydrostatic pressure
- C. Decreased glomerular oncotic pressure
- D. Efferent arteriolar vasoconstriction
- E. Afferent arteriolar vasoconstriction

Answer: E. Afferent arteriolar vasoconstriction

Explanation:

- A: Favors filtration because increased glomerular hydrostatic pressure increases GFR.
- B: Decreased Bowman's capsule pressure increases net filtration.
- C: Decreased plasma oncotic pressure increases filtration.
- D: Moderate efferent constriction can increase GFR, although severe constriction can eventually reduce GFR.
- E: Afferent constriction decreases glomerular capillary pressure and decreases GFR.

55. Secretion occurs in which renal tubular segments?

- A. PCT, DCT, collecting tubule
- B. PCT, descending loop, DCT
- C. PCT, ascending loop, DCT
- D. PCT, ascending loop, collecting tubule
- E. Throughout all segments

Answer: A. PCT, DCT and collecting tubule

Explanation:

- A: Important sites for secretion of various substances.
- B: Descending limb is mainly water-permeable and is not a major secretory segment.
- C: Ascending limb is not a major general secretion site.
- D: Less complete than A for the intended standard description.
- E: Secretion is not uniform throughout the nephron.

56. Substances most often actively secreted by kidneys include:

- A. Water, Na^+ , Cl^- , K^+ , H^+
- B. Water, Na^+ , Cl^- , K^+
- C. Na^+ , Cl^-
- D. K^+ and H^+
- E. Na^+ , Cl^- , K^+ , H^+

Answer: D. K^+ and H^+

Explanation:

- Important tubular secretions include K^+ , H^+ , NH_4^+ , creatinine, and organic acids/bases.
- A: Water, Na^+ and Cl^- are mainly reabsorbed, not secreted.
- B: Same problem.
- C: Na^+ and Cl^- are predominantly reabsorbed.
- D: Correct.
- E: Na^+/Cl^- are not typically listed as major actively secreted substances.

57. Tubular secretion makes kidneys busier than tubular reabsorption.

- A. True
- B. False

Answer: B. False

Explanation:

- Tubular reabsorption is quantitatively much greater than tubular secretion.
- The kidneys filter a huge amount of plasma and then reabsorb most of the filtered water and solutes.

58. Brain structure responsible for hunger and satiety:

- A. Medulla
- B. Hypothalamus
- C. Pons
- D. Thalamus
- E. Cerebellum

Answer: B. Hypothalamus

Explanation:

- A: Major autonomic/respiratory/cardiovascular functions.
- B: Major center for appetite and energy balance.
- C: Bridge and respiratory functions.
- D: Sensory relay.
- E: Coordination and balance.

59. Essential stimulus for secretion of secretin:

- A. Gastrin
- B. Histamine
- C. Somatostatin
- D. Acidic chyme in duodenum

E. Fat digestion products

Answer: D. Acidic chyme in duodenum

Explanation:

- Low duodenal pH stimulates S cells to release secretin, which causes pancreatic/biliary bicarbonate secretion.
- A: Gastrin stimulates gastric acid secretion.
- B: Histamine stimulates gastric HCl through H₂ receptors.
- C: Somatostatin is an inhibitory hormone.
- D: Correct.
- E: Fatty acids are especially important for CCK; acid is the classic stimulus for secretin.

60. GIT layer containing many glands in the small intestine:

- A. Serosa
- B. Muscularis
- C. Submucosa
- D. Mucosa
- E. All

Answer: D. Mucosa

Explanation:

- Brunner glands are located in the submucosa of the duodenum.
- A: Outer covering.
- B: Responsible mainly for motility.
- C: Contains Brunner glands.
- D: Intestinal crypts/glands are located in mucosa, so the wording is ambiguous.
- E: Not all layers contain glands.

61. Center for chewing is located in:

- A. Medulla
- B. Pons
- C. Midbrain
- D. Spinal cord
- E. Cerebrum

Answer: B. Pons

Explanation:

- The chewing/mastication pattern generator is associated with the pons, with involvement of trigeminal motor pathways.
- A: Incorrect.
- B: Correct.
- C: Incorrect.
- D: Incorrect.
- E: Voluntary cortical control contributes but the central chewing pattern generator is in the brainstem.

62. Which food item is chemically digested in the stomach?

- A. Protein
- B. Fat
- C. CHO
- D. Vitamin
- E. Mineral

Answer: A. Protein

Explanation:

- Pepsin begins significant protein digestion in the stomach.
- A: Main chemical digestion in stomach.
- B: Some gastric fat digestion occurs via gastric lipase, but protein is the classic answer.
- C: Salivary amylase may continue briefly, but acidic gastric conditions eventually inhibit it.
- D: Vitamins are absorbed rather than digested.
- E: Minerals are not chemically digested.

71. Identify a steroidal hormone:

- A. Insulin
- B. ADH
- C. Growth hormone
- D. Aldosterone
- E. Oxytocin

Answer: D. Aldosterone

Explanation:

- Aldosterone is synthesized from cholesterol in the adrenal cortex.
- A: Peptide hormone.
- B: Peptide hormone.
- C: Protein hormone.
- D: Steroid hormone.
- E: Peptide hormone.

72. Which is a neurohormone?

- A. Growth hormone
- B. Prolactin
- C. Parathormone
- D. ADH
- E. Cortisol

Answer: D. ADH

Explanation:

- ADH is synthesized by hypothalamic neurons and released from the posterior pituitary.
- A: Anterior pituitary hormone.
- B: Anterior pituitary hormone.
- C: Parathyroid hormone.
- D: Neurohormone.
- E: Adrenal cortex steroid.

73. Non-steroidal hormone that can diffuse through lipid bilayer:

- A. GH
- B. Epinephrine
- C. T3
- D. ADH
- E. Insulin

Answer: C. T3

Explanation:

- T3 is not a steroid, but it is lipid-soluble and can cross cell membranes.

- A: GH is a peptide/protein; uses membrane receptor.
- B: Epinephrine is a catecholamine; water-soluble.
- C: T3 is a thyroid hormone, derived from tyrosine but lipid-soluble.
- D: ADH is a peptide.
- E: Insulin is a peptide.
- High-yield exception: Steroid and thyroid hormones use intracellular receptors.

74. Mechanism of action of ADH:

- A. Gene expression
- B. Second messenger formation
- C. Neural pathway
- D. Ion channel opening

Answer: B. Second messenger formation

Explanation:

- ADH acts through different receptors.
- V_2 receptor $\rightarrow$ Gs $\rightarrow$ adenylyl cyclase $\rightarrow$ cAMP $\rightarrow$ PKA $\rightarrow$ aquaporin-2 insertion.
- A: This is more characteristic of intracellular/nuclear receptor hormones.
- B: V_2 receptor uses cAMP.
- C: ADH is a hormone, not a neural pathway mechanism.
- D: Not the primary mechanism.

75. Determinant factor for fetus to become male:

- A. Presence of hypothalamus
- B. Presence of penis
- C. Presence of Y chromosome
- D. Absence of X chromosome
- E. All

Answer: C. Presence of the Y chromosome

Explanation:

- More precisely, the key determinant is the SRY gene on the Y chromosome, which initiates testis development.
- A: Incorrect.
- B: Penis develops as a consequence of male differentiation.
- C: Correct.
- D: Males normally have an X chromosome: XY.
- E: Incorrect.

76. Müllerian duct develops into:

- A. Ovaries
- B. Female genital tract
- C. Male accessory glands
- D. Female external genitalia
- E. Ovaries

Answer: B. Female genital tract

Explanation:

- Müllerian/paramesonephric ducts form the fallopian tubes, uterus, cervix, and upper vagina.
- A: Ovaries develop from gonadal ridges, not Müllerian ducts.
- B: Correct.

- C: Male accessory structures mainly derive from Wolffian duct/other structures.
- D: External genitalia have different embryologic origins.
- E: Duplicate of A.

77. Erectile phase of male sexual act is controlled by:

- A. Sympathetic NS
- B. Parasympathetic NS
- C. Somatic NS
- D. Testosterone
- E. FSH

Answer: B. Parasympathetic NS

Explanation:

- Point and shoot mnemonic: Point = Parasympathetic (erection). Shoot = Sympathetic (ejaculation).
- Parasympathetic activity → NO → increased cGMP → relaxation of penile vascular smooth muscle → erection.
- A: Sympathetic is important for ejaculation.
- B: Correct.
- C: Somatic pathways participate in contraction but are not the primary erectile mechanism.
- D: Testosterone supports libido/sexual function but does not directly cause erection.
- E: FSH is involved in spermatogenesis.

78. Hormone of ovulation:

- A. FSH
- B. Estradiol
- C. Progesterone
- D. hCG
- E. LH

Answer: E. LH

Explanation:

- LH surge triggers ovulation.
- A: Stimulates follicular development.
- B: High sustained estradiol triggers positive feedback leading to LH surge.
- C: Dominant after ovulation.
- D: Maintains corpus luteum during early pregnancy.
- E: LH surge triggers ovulation.

79. Usual site of fertilization:

- A. Ovary
- B. Fallopian tubes
- C. Uterus
- D. Vagina
- E. Clitoris

Answer: B. Fallopian tubes

Explanation:

- More specifically, the ampulla of the uterine tube is the usual site.
- A: Ovulation occurs here, not normally fertilization.
- B: Correct.
- C: Implantation normally occurs in uterus.

- D: Incorrect.
- E: Incorrect.

80. XXY infertile male:

- A. Turner syndrome
- B. Klinefelter syndrome
- C. XXXYY syndrome
- D. Trisomy
- E. Cushing syndrome

Answer: B. Klinefelter syndrome

Explanation:

- Klinefelter = 47,XXY.
- Typical features: Male phenotype, small firm testes, hypogonadism, infertility, often tall stature, gynecomastia may occur.
- A: Turner is 45,X female phenotype.
- B: Klinefelter is 47,XXY male.
- C: Not the classic syndrome described.
- D: Trisomy is a general chromosomal abnormality, not the specific diagnosis.
- E: Endocrine disorder due to excess cortisol.

